EXCLUSIVE

IAS PRELIMS 2026

CURRENT AFFAIRS

GS-1 (DYNAMIC)

REVISION

MODULE - 23

SCIENCE & TECHNOLOGY-6

HOW DO YOU DIFFERENTIATE BHARAT SMALL REACTOR (BSR) AND BHARAT SMALL MODULAR REACTOR (BSMR)?

- Bharat Small Reactors refer to incrementally modified forms of India's existing Pressurized Heavy Water Reactors (PHWR).
- The majority of the 22 operational indigenously built nuclear reactors have a capacity rating of 220 MW. India also has experience in building an 85 MW reactor for its nuclear submarine.
- Bharat Small Modular Reactor (BSMR) is a nascent technology being researched globally.
- In the 2024-25 Budget speech in July, the Finance Minister said that the government would be partnering with the private sector for setting Bharat Small Reactors, Research and Development of Bharat Small Modular Reactor, and research and development of New Age Technologies for nuclear energy.
- There are several advantages with Small Modular Reactors. These are:
  - ✓ Speed of installation.
  - ✓ Cost-saving.
  - ✓ Potential enhanced safety.
  - ✓ Flexibility of installation.

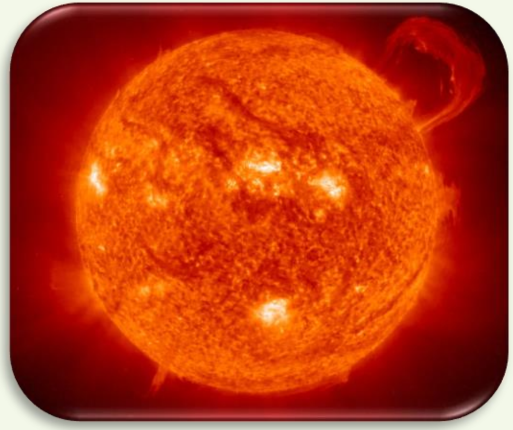
Some of the developments on the Nuclear Energy front

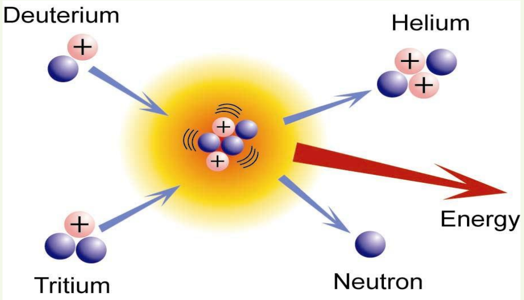
- The Government accorded in-principle approval to set up 6 x 1208 MW nuclear power plant in cooperation with the USA at Kovvada in Andhra Pradesh.
- A significant discovery of new deposit in India's oldest Uranium Mine, the Jaduguda Mines, have been made in and around the existing mine.
- First two units of India's indigenous 700 MW PHWRs at the Kakrapar, Gujarat began commercial operations.
- NPCIL and National Thermal Power Corporation (NTPC) have signed a supplementary Joint Venture Agreement to develop nuclear power facilities in the country. The Joint Venture is named "ASHVINI" will function within the existing legal framework of the Atomic Energy Act 1962. They will build, own, and operate nuclear power plants, including the upcoming 4x700 MWe PHWR Mahi-Banswara Rajasthan Atomic Power Project.

A kilogram of fusion fuel contains about 10 million times as much energy as a kilogram of coal or oil or gas. ITER is expected to begin deuterium-tritium fusion reactions by 2039, producing 500 MW of fusion power.

PRIME MINISTER's VISIT TO ITER IN FRANCE

- Prime Minister visited International Thermonuclear Experimental Reactor or ITER with French President Emmanuel Macron in Cadarache, Southern France.
- Modi's visit to ITER facility marks a first by any head of state or head of government to ITER.
- The ITER Project is a collaboration between the European Union, the United States, China, India, Japan, Russia and South Korea to build a prototype fusion reactor in Southern France with the World's largest Tokamak, an experimental device. It is to demonstrate the scientific and technological feasibility of fusion energy.
- Joint European Torus (JET) in the UK, Tokamak 60 (JT-60) of the Japan Atomic Energy Research Institute, Tokamak Fusion Test Reactor (TFTR) at the Princeton Plasma Physics Laboratory in New Jersey, US are some of the Nuclear Fusion research hubs in the World.





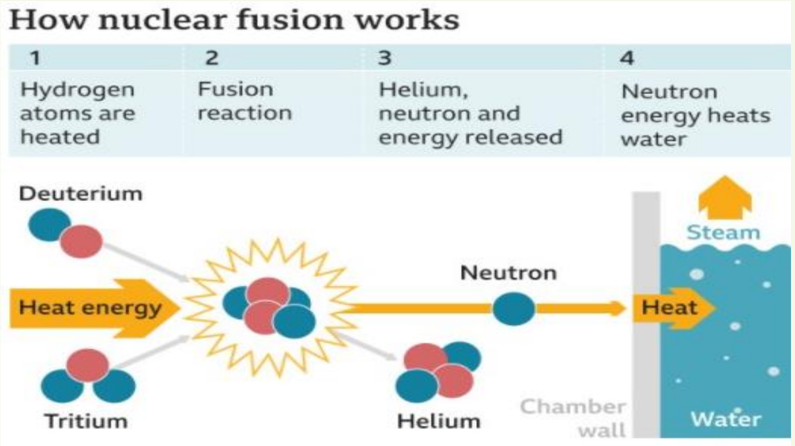
WHY IS IT DIFFICULT FOR GETTING ENERGY THROUGH NUCLEAR FUSION?

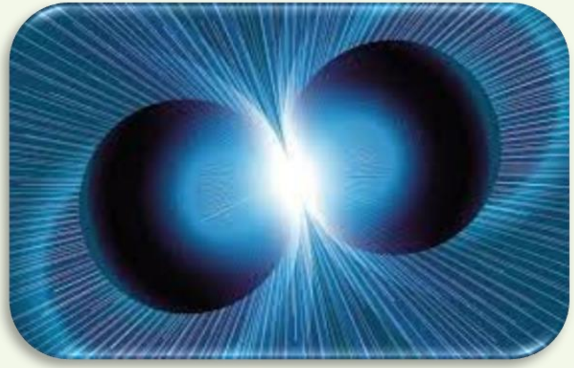
- Normally, atomic nucleii repel each other, if we try to bring them closer. To force them to come close and ultimately fuse together, special conditions have to be generated in the form of very high pressure and extremely high temperatures. We are finding it difficult to attain such situation practically.
- Stars like the Sun radiate huge quantities of energy because of the nuclear fusion reaction taking place inside their core - hydrogen is continuously changing to helium.

So, the basic requirement for a fusion reaction is to create a star-like situation inside the reactor in terms of temperature and pressure.

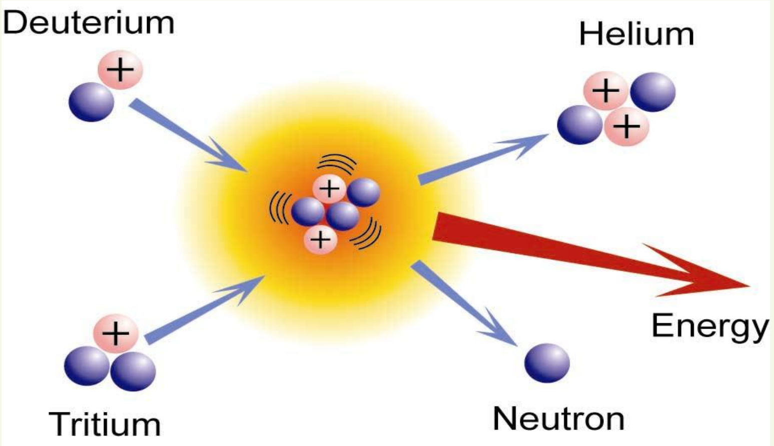
WHAT ARE THE COMPLEXITIES IN DEVELOPING NUCLEAR FUSION ENERGY?

Learning Space





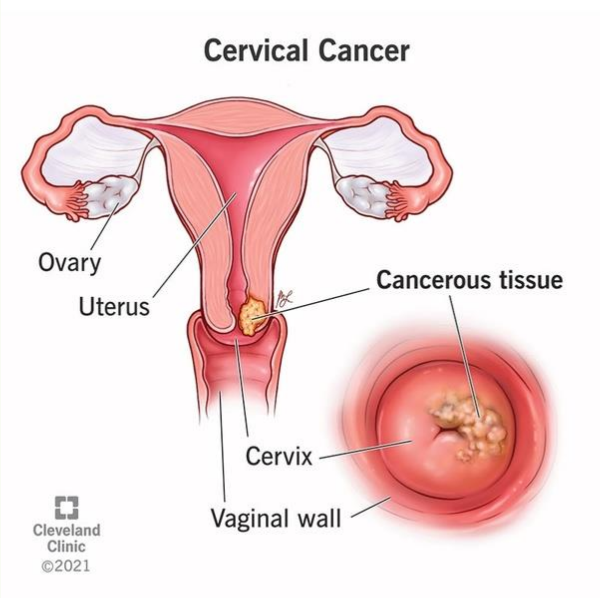
- The pressures that are possible on Earth are much lower in comparison to the Sun. Hence, temperatures to produce fusion need to be much higher, say above 100 million Celsius. At such high temperatures, a fully ionized gas plasma is formed. The plasma will then be ignited to create fusion.
- No materials exist that can withstand direct contact with such heat. Scientists are looking at the possibilities of super heated gas / plasma held inside a doughnut shaped magnetic field. Hence, Plasma is held in place by using superconductor electromagnets and as it spins around, it fuses and releases tremendous energy as the heat.



WHAT IS ITER-INDIA?

- ITER-India is a special project under Institute for Plasma Research (IPR). It is governed by the Empowered Board, which is chaired by the Secretary, Department of Atomic Energy (DAE).
- India became a full seventh partner of ITER in December 2005.
- Institute for Plasma Research (IPR) is located in Gandhinagar, Gujarat and is the Indian Domestic Agency to design, build and deliver the Indian in-kind contribution to ITER.
- Around 200 Indian scientists, associates, and notable industry players such as L&T, Inox India, TCS, TCE, HCL Technologies, are associated with the ITER project.

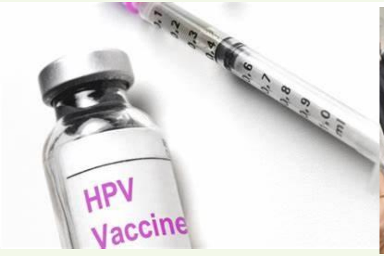
What do you understand by Cervical Cancer?
Why is it considered a major problem for Indian Women?



- Cancer of the Cervix (literally, the neck of the womb) is unique among cancers because almost 99% of the cases are linked to infection with the Human Papillomavirus (HPV), a common virus transmitted through sexual contact.
- While most HPV infections resolve spontaneously and the women remain symptom-free, persistent infection can lead to cervical cancer.
- It is the second leading cause of cancer-related deaths among women in India. Annually, more than 77,000 deaths are happening.
- It is the second most frequent cancer among Indian women between 15 and 44 years, after breast cancer.

There is availability of Vaccine. The screening for Cervical Cancer is just under 2%.

What is the Two-pronged Strategy needed to control Cervical Cancer in India?



The above-mentioned two-pronged strategy is needed to control cervical cancer for women in India. India accounts for around 25% of the cervical cancer deaths in the World.

1

- Vaccination for girls aged 9 to 14 was announced in the Interim Budget 2024-25. The National Technical Advisory Group on Immunization recommended the introduction of HPV Vaccine in the Universal Immunization with a one time catch up for 9–14-year-old adolescent girls followed with routine introduction at nine years.
- Cervavac Vaccine manufactured by Serum Institute of India is available at Rs. 2,000 per dose. Other Vaccine Gardasil 4 (quadrivalent vaccine) is currently priced at Rs 3,927 per dose.

2

- The other important aspect is simple screening in a public health setting with minimal tools - the human eye, a dilution of white vinegar and a dab of Lugol’s Iodine - can identify cervical cancer in the early stages.
- If any abnormal growth is there, a simple procedure - cryotherapy can be done while the patient is awake to destroy the abnormal growth.

What do you understand by Mission Divyastra?

- India joined an elite club of nations on March 11, 2025 with the first successful flight test of an Agni-5 Ballistic Missile armed with Multiple Independently Targetable Re-entry Vehicles (MIRVs).
- Only the permanent members of the United Nations Security Council -- the United States, the United Kingdom, Russia, China, and France are confirmed to have operational missiles that use MIRVs.
- With MIRV Technology, each warhead is carried in a separate re-entry vehicle and can be programmed to hit a separate target.
- A missile armed with MIRVs can release its warheads at different speeds and even in different directions.
- The flight test of India's MIRV Technology was carried out by the DRDO from Dr. APJ Abdul Kalam Island in Odisha.

Why is MIRV success a major milestone?

- MIRVs make defending the intended targets much more difficult for the adversary nation. These are effective Ballistic Missile Defence counter measures.
- This technology makes an Agni-5 Missile more difficult to intercept because its warheads will approach their intended targets along with the mother vehicles and multiple decoys.
- Multiple missiles armed with MIRVs could even make a comprehensive BMD system cost-prohibitive for the adversary.
- Given that China has BMD capabilities, MIRVs will boost the Agni-5's chances of hitting its targets successfully.

What is Copernicus programme?
What are Sentinel group of satellites?

- Copernicus is European Union's Earth observation programme.
- This flies a series of satellite sensors called Sentinels, which monitor everything from damage wrought by earthquakes to the health of staple food crops.

SENTINEL GROUP OF SATELLITES

- Sentinel-1 ... Radar satellite that can see the Earth's surface in all weathers
- Sentinel-2 ... Multi-wavelength detectors to study principally land changes
- Sentinel-3 ... To observe ocean properties and behaviour
- Sentinel-4 ... To measure atmospheric gases
- Sentinel-5 ... To help monitor air quality
- Sentinel-6 ... Sea-surface height series

What is latency? How 5G reduces latency?

- In computer networking, latency is an expression of how much time it takes for a data packet to travel from one designated point to another. In other words, latency is the time a device takes to communicate with the network.
- The latency stands at 50 milliseconds for 4G networks across the world. With 5G, latency came down to between 1 to 10 milliseconds. 5G will also have faster speeds.

UNDERSTAND BRIEFLY!

- 1G ... only mobile voice calls
- 2G ... mobile voice calls and sending of short text messages.
- 3G ... web browsing on mobile devices
- 4G .... speed and latency improved

A BRIEF ON STARLINK

- Starlink is a Low-Earth Orbit (LEO) satellite constellation with over 7,000 satellites that provide internet access to users with ground terminals. It is already selling in around 40 countries.
- These satellites constantly orbit the Earth blanketing practically its entire habitable surface with coverage as long as a given terminal on the ground has visibility to the sky.
- Satellites work in conjunction with ground stations on Earth, which are physically connected to the Internet like any other network and beam up connections wirelessly to satellites above them at any given point.

Starlink noted on its website that it is a constellation of thousands of satellites that orbit Earth at around 550 km.

BOTH THESE COMPANIES HAD TIE UPS!

AIRTEL WITH OneWeb

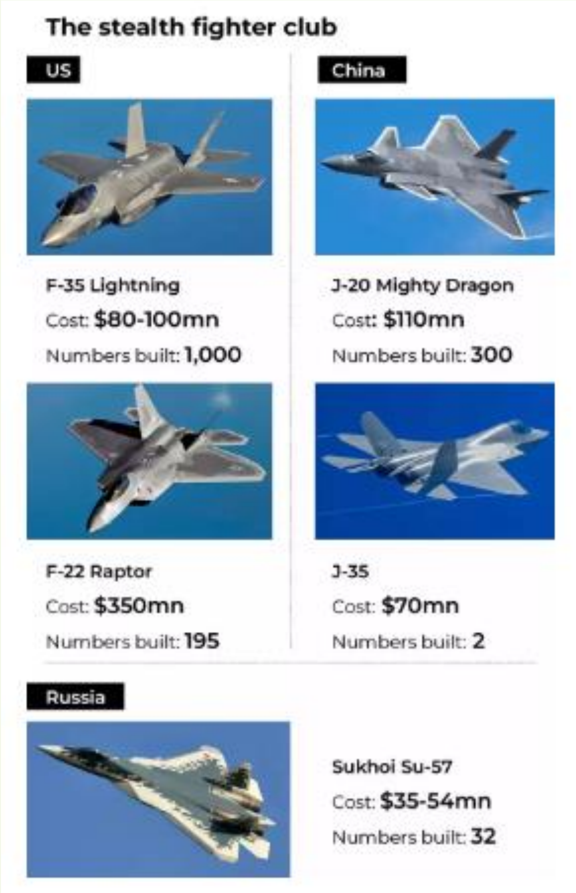
- Airtel has existing alliance with Eutelsat OneWeb, which also provides satellite connectivity.
- There are more than 630 satellites along 12 carefully synchronized orbital planes, 1,200 km above in Low Earth Orbit (LEO).

JIO WITH SES

- In 2023, Jio introduced its own JioSpaceFiber satellite internet service. It stated that it was partnering with SES to access the world's latest in Medium Earth Orbit Satellite Technology.
- SES has 70 satellites operating in two different orbits and broadcasts to over 1 billion TV viewers worldwide.

WHAT DO YOU UNDERSTAND BY A STEALTH FIGHTER?

- These are mostly fifth generation fighter jets designed to avoid detection by radar and other electronic means. They use a variety of technologies to reduce the radar signature - infrared, visible light, radio frequency and audio emissions.
- Stealth aircraft avoids sharp curves, right angles and large surfaces. Instead, they use curved surfaces with large radii and many small flat plains. They are made of special radar absorbent materials.



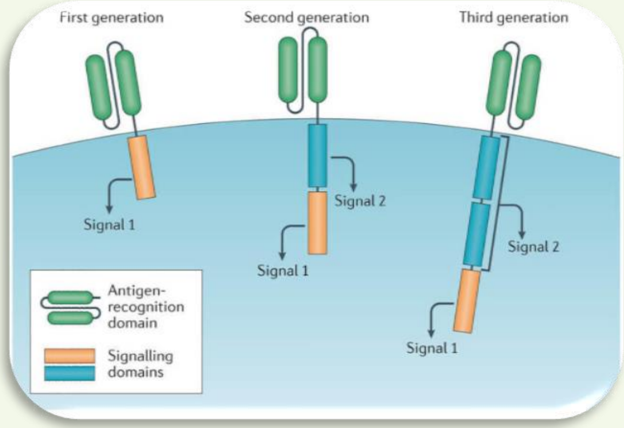
CHINA IS MUCH AHEAD IN AIR POWER IN COMPARISON TO INDIA

The Indian Air Force (IAF) currently operates only 31 squadrons (each with 18 aircraft) far below the sanctioned strength of 42. The shortfall is largely due to IAF's inability to replace the ageing Soviet-era fleet, especially the accident-prone MiG-21s. The Indian Air Force (IAF) officially retired its final MiG-21 fighter jets on September 26, 2025,

India vs China air power

PLAAF	IAF
2nd largest in world	4th largest in world
2500 aircraft	900 aircraft
1200 fighters	600 fighters
Has 5th Gen Stealth Jets	No 5th Gen Stealth Jets
Has Long Range Strategic Bomber Fleet	No Long Range Strategic Bomber Fleet
More AWACS and Combat Drones	Less AWACS and Combat Drones

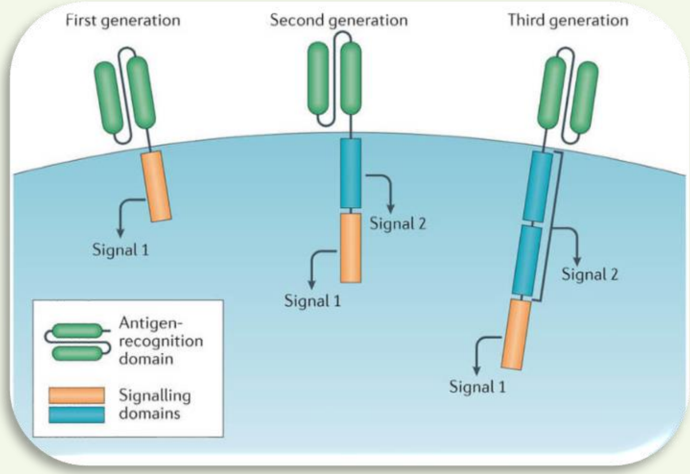
WHAT DO YOU UNDERSTAND BY CAR T-CELL THERAPY?



- CAR T-Cell Therapy is a cell therapy in which a part of the body's immune system, T-Cells are primed to target and destroy cancer cells. Scientists focused on T-Cells because they are genetically programmed to attack foreign particles and with bioengineering, they can become more precise and potent warriors.
- CAR-T cells belong to a branch of oncology therapeutics called immunotherapy, which is now hailed as the fifth pillar of cancer therapy after surgery, chemotherapy, radiation and targeted therapy.
- T-Cells are taken from patients' blood, these cells are genetically engineered in a laboratory to produce special structures called Chimeric Antigen Receptors (CARs).
- Receptors can recognize and bind to specific proteins, i.e., particular antigens on the surface of cancer cells. CAR T-Cells circulate throughout the body and as per their programming, recognize and attach to cancer cells.

However, there are challenges like their response against solid tumors, side effects, long term survival of CAR T-Cells in the body.

WITH THE INSERTION OF NEW GENETIC MATERIAL, RISK OF INSERTIONAL MUTAGENESIS ALWAYS EXIST!



- The process of creating CAR-T Cells involves using viruses to insert new genetic material into T-Cells. These viruses called retroviruses, are engineered to carry the gene for a Chimeric Antigen Receptor (CAR) into the T-Cell.
- While these retroviruses are modified to be safe, there is always a risk that the new genetic material could disrupt other important genes and potentially lead to cancer, a phenomenon known as insertional mutagenesis. This means a new genetic material is added to a cell.

WHAT ARE iVDPV AND cVDPV?

IMMUNODEFICIENCY VDPV

- When Oral Polio Vaccine is administered to a child with weak immune system, the virus might, in certain extreme cases, genetically mutate in the intestine and cause polio. This is called Immunodeficiency VDPV.
- These are usually observed in children with primary immunodeficiency. Such children are unable to mount an immune response and unable to clear the intestinal vaccine virus infection.

World Health Organization

CIRCULATING VDPV

Mutated virus, when excreted by the child, can lead to community spread, this is called circulating Vaccine Derived Poliovirus.

HOW TO CONTAIN VDPV?

- WHO suggests the need to phase out OPV usage in all countries that eliminated wild polio virus to avoid the risk of VDPV and instead shift to Inactivated Polio Vaccine to maintain immunity levels in the population.
- IPV is costlier and requires a trained health worker to be administered.

WE EXPRESS OUR SINCERE THANKS FOR VIEWING THIS VIDEO

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